PROJECT 9

INTELLIGENT PROPERTY PRICE PREDICTION & AGENTIC REAL ESTATE ADVISORY

From Market Analytics to Autonomous Investment Advice

PROJECT OVERVIEW

This project involves the design and implementation of an **AI-driven real estate analytics system** that predicts property prices and evolves into an agentic AI real estate advisory assistant.

In **Milestone 1**, the system applies classical machine learning techniques to historical listing data and location attributes to predict property values and analyze market drivers.

In **Milestone 2**, the same system is extended into an agent-based AI application that autonomously reasons about property characteristics, retrieves market insights, and generates investment recommendations.

The project emphasizes real-world real estate decision-making, progressive development from predictive analytics to agentic AI workflows, and public deployment.

CONSTRAINTS & REQUIREMENTS

TEAM SIZE **3–4 Students**	**API BUDGET** **Free Tier Only**
FRAMEWORK	**HOSTING**

LangGraph (Recommended)	**Mandatory**

APPROVED TECHNOLOGY STACK

LLMS (MILESTONE 2)

- Open-source models
- Free-tier APIs

AGENT FRAMEWORK

- LangGraph (Recommended)
- Chroma / FAISS (RAG)

ML MODELS (MILESTONE 1)

- Linear Regression
- Random Forest / Decision Trees
- Scikit-Learn (Pipelines)

UI FRAMEWORK

- Streamlit
- Gradio

HOSTING PLATFORM (MANDATORY)

- Hugging Face Spaces
- Streamlit Community Cloud
- Render (Free Tier)

WARNING: Localhost-only demonstrations will not be accepted.

MILESTONE 1: ML-BASED PROPERTY PRICE PREDICTION

MID-SEM SUBMISSION

Objective: Design and implement a machine learning-based property price prediction system that estimates values using historical listing data and features. Focus on classical ML workflows *without LLMs*.

Functional Requirements:

- Accept property feature data as input.
- Perform data preprocessing (Encoding, Scaling).
- Predict property prices or price ranges.
- Display predictions through a user interface.

TECHNICAL REQUIREMENTS (ML)

- **Preprocessing:** Missing values, Categorical encoding.
- **Models:** Linear Regression, Random Forests.
- **Evaluation:** MAE, RMSE, R-squared.
- **Features:** Location, Size, Rooms, Amenities.

INPUTS & OUTPUTS

- **Input:** Property CSV data.
- **Output:** Price estimate/range.
- **Metrics:** Price driver analysis.

MID-SEM DELIVERABLES

- Problem understanding & Real Estate Use-case.
- Input–output specification.
- System architecture diagram.
- Working local application with UI.
- Model performance evaluation report.

MILESTONE 2: AGENTIC AI REAL ESTATE ADVISORY ASSISTANT

END-SEM SUBMISSION

Objective: Extend the price prediction system into an agentic AI real estate advisory assistant that autonomously reasons about property value, retrieves market trends, and generates investment recommendations.

Functional Requirements:

- Accept property details and user preferences.
- Analyze predicted prices and market conditions.
- Retrieve real estate trends and regulations.
- Generate structured advisory reports for investors.

TECHNICAL REQUIREMENTS (AGENTIC)

- **Framework:** LangGraph (Workflow & State).
- **RAG:** Retrieval of market insights (Chroma/FAISS).
- **State:** Explicit state management across steps.
- **Prompting:** Strategies to reduce hallucinations.

STRUCTURED OUTPUT

- **Summary:** Property Valuation & Market view.
- **Comps:** Comparable property analysis.
- **Action:** Buy/Invest recommendation.
- **Disclaimer:** Financial/Legal notices.

END-SEM DELIVERABLES

- Publicly deployed application.
- Agent workflow documentation.
- Structured advisory report.
- GitHub Repository & Complete Codebase.
- Demo Video (Functionality walkthrough).

Final Artifacts: Hosted Link, GitHub Repo, Demo Video.

EVALUATION CRITERIA

PHASE	WEIGHT	CRITERIA
Mid-Sem (Milestone 1)	25%	• Correct application of ML techniques • Quality of Preprocessing & Feature Engineering • Appropriateness of evaluation metrics • UI Usability & Code modularity
End-Sem (Milestone 2)	30%	• Quality of Agentic Reasoning & Decision Support • Correct RAG integration & State management • Utility of Advisory Insights • Responsible AI & Deployment Quality